

Back of the Envelope Estimation

Before designing the architecture, we estimate:

- Scale
- Traffic
- Storage
- Bandwidth
- Infrastructure cost

These rough calculations help us:

None

Understand bottlenecks and system requirements

Assumptions

The following assumptions are provided during the interview.

Metric	Value
Daily Active Users (DAU)	5 Million
Videos watched per user/day	5
Uploading users	10%
Videos uploaded/user/day	1

Average video size	300 MB
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1. Daily Video Uploads

Uploading Users

Only:

None

10% of users upload videos

So:

$$5,000,000 * 10\% = 500,000$$

Total uploads/day:

None

500,000 videos/day

2. Daily Storage Requirement

Average video size:

None

300 MB

Total storage/day:

$$500,000 * 300\text{MB} = 150,000,000 \text{ MB}$$

Convert to TB:

$150,000,000 / 1024 / 1024 \approx 150\text{TB}$

Final Result

None

$\approx 150\text{ TB of new storage required per day}$

3. Total Videos Watched Per Day

Each user watches:

None

5 videos/day

Total views/day:

$5,000,000 * 5 = 25,000,000$

Final Result

None

$25\text{ million video streams/day}$

4. Daily Streaming Bandwidth

Average video size:

None

0.3 GB

Total outbound data/day:

$25,000,000 * 0.3\text{GB} = 7,500,000 \text{ GB/day}$

Final Result

None

≈ 7.5 million GB/day

This is:

- Massive internet traffic
- Extremely bandwidth intensive

5. CDN Cost Estimation

Video streaming platforms heavily depend on:

None

CDN (Content Delivery Network)

because videos must be delivered:

- Quickly
- Globally
- With low latency

CloudFront Pricing Assumption

Assume:

- Using Amazon CloudFront
- All traffic served within the US
- Average transfer cost:

None

\$0.02 per GB

Daily CDN Cost

$7,500,000 \text{ GB} * \$0.02 = \$150,000/\text{day}$

Final Result

None

$\approx \$150,000 \text{ per day}$

Yearly CDN Cost

$150,000 * 365 \approx \$54.75 \text{ million/year}$

Final Result

None

$\approx \$55 \text{ million/year}$

Key Observation

From the calculation:

None

CDN bandwidth is one of the biggest expenses

for video platforms.

Even though large companies negotiate lower CDN pricing:

- The cost still remains enormous.
- Bandwidth optimization becomes critical.

Why These Estimates Matter

These estimations reveal major engineering priorities.

Problem	Importance
Storage growth	Very high
CDN cost	Extremely high
Video bandwidth	Massive
Scalability	Essential
Compression optimization	Critical
Caching	Mandatory

Important Insights

1. Storage Grows Extremely Fast

Daily:

None

150 TB/day

Yearly:

$150 * 365 \approx 54,750 \text{ TB} \approx 54.7 \text{ PB/year}$

The platform needs:

- Distributed object storage
- Replication
- Cold storage policies

2. CDN Optimization is Critical

Possible optimizations:

- Cache popular videos aggressively
- Adaptive bitrate streaming
- Compress videos
- Regional CDN placement
- Edge caching

3. Video Compression is Necessary

Without transcoding/compression:

- Costs explode
- Bandwidth usage becomes unsustainable

Thus:

None

Video transcoding pipeline is mandatory

4. Scalability Must Be Distributed

Single server solutions are impossible.

The system requires:

- Distributed storage
- Distributed CDN
- Distributed transcoding workers
- Horizontal scaling

Final Understanding

The back-of-the-envelope calculations help us understand that:

None

Video streaming systems are primarily bandwidth and storage problems

The two biggest infrastructure costs are:

1. Video storage
2. CDN delivery

These estimations guide the architecture decisions in the next design steps.